

$$y_i^* | \alpha, x_i^*, \beta, \sigma^2 \stackrel{\text{ind}}{\sim} \mathcal{N}(\alpha + x_i^{*\top} \beta + \phi_i + \sigma_i, \sigma^2) \quad i = \overline{1, n}$$

$$\alpha \sim \mathcal{N}(0, \underline{1})$$

$$\beta \sim \mathcal{N}(0, \underline{I}_p)$$

$$\underline{\phi} | W, \tau^2, \rho \sim \mathcal{N}_n(\underline{0}, \tau^2 [\rho(\underline{0} - W) + (1-\rho)\underline{I}]^{-1})$$

$$\tau^2 \sim \text{inv-gamma}(1, 0.01)$$

$$\rho \sim U(0, 1)$$

$$\rightarrow \underline{\phi} | W, \tau^2, \rho \sim \mathcal{N}_n(\underline{0}, \underbrace{\tau^2 [\rho(\underline{0} - W) + (1-\rho)\underline{I}]^{-1}}_Q)$$

$$\tau^2 \sim \text{gamma}(1, 0.01)$$

⊗ Deal with sparse matrix

$$\mathcal{Q}(\underline{\phi} | W, \tau, \rho) = \frac{1}{|\underline{Q}^{-1}|^{1/2}} (2\pi)^{-N/2} \exp \left\{ -\frac{1}{2} \underline{\phi}^\top \underline{Q} \underline{\phi} \right\}$$

$$= (2\pi)^{-N/2} |\underline{Q}|^{1/2} \exp \left\{ -\frac{1}{2} \underline{\phi}^\top \underline{Q} \underline{\phi} \right\}$$

$$\Rightarrow \log(\pi(\underline{\phi} | W, \tau, \rho)) = \boxed{\frac{-N}{2} \log(2\pi)} + \frac{1}{2} \log |\underline{Q}| - \frac{1}{2} \underline{\phi}^\top \underline{Q} \underline{\phi}$$

Substitute $\underline{Q} = \tau^2 [\rho(\underline{0} - W) + (1-\rho)\underline{I}]$

$$\Rightarrow \dots \propto \frac{1}{2} \log \underbrace{|\tau^2 \underline{V}|}_{\text{det}} - \frac{\tau^2}{2} \underline{\phi}^\top \underbrace{\underline{V} \underline{\phi}}_{\checkmark}$$

$$\propto \boxed{\frac{\overset{\text{det}}{\cancel{N}}}{2} \log \tau^2} + \frac{1}{2} \log |V|$$

$$\propto \frac{1}{2} \log |p(D-W) + (1-p)I_N| \xrightarrow{\text{det}}$$

$$+1) |p(D-W) + (1-p)I_N| = \prod_{i=1}^N (p \lambda_i + (1-p)) \quad (*)$$

$\xrightarrow{\text{eigenvalues of } D-W}$

$$\Rightarrow \dots \propto \frac{1}{2} \log \left(\prod_{i=1}^N (p \lambda_i + (1-p)) \right)$$

$$\propto \boxed{\frac{1}{2} \sum_{i=1}^N \log (p \lambda_i + (1-p))}$$

$$\Rightarrow \log (\pi(\underline{\phi} | W, \tau, p)) \propto \frac{N}{2} \log(\tau^2) + \frac{1}{2} \sum_{i=1}^N \log(p \lambda_i + (1-p)) - \frac{\tau^2}{2} \underline{\phi}^T V \underline{\phi}$$

$$\propto \frac{1}{2} \left[N \log(\tau^2) + \sum_{i=1}^N \log(p \lambda_i + (1-p)) - \tau^2 \underline{\phi}^T V \underline{\phi} \right]$$

in which λ_i are eigenvalues of $D-W$
 $V = p(D-W) + (1-p)I$

$$\propto \frac{1}{2} \left[N \log(\tau^2) + \sum_{i=1}^N \log(p \lambda_i + (1-p)) - \tau^2 (p \underline{\phi}^T (D-W) \underline{\phi} + (1-p) \underline{\phi}^T \underline{\phi}) \right]$$

(*) Explanation

1. W : adjacency matrix $\Rightarrow$ symmetric
 $D = [d_{ij}] \Rightarrow \text{diagonal} \Rightarrow \text{symmetric}$

$$\left\{ \begin{array}{l} d_{ii} = \sum_{j=1}^N W_{ij} \\ d_{ij} = 0 \quad \forall i \neq j \end{array} \right.$$

$\Rightarrow D-W$ symmetric

2. Every real symmetric $A \Rightarrow$ spectral decomposition

$$A = U \Lambda U^T$$

+) $U^T U = I$

+) $\Lambda = \text{diag}(\lambda_1 \dots \lambda_N)$ (eigenvalues of A)

3. $U^T U = I \Rightarrow |U \Lambda U^T| \xrightarrow{\text{cyclic}} |U^T U A| = |A|$

+) Apply: $V = p(D-W) + (1-p)I$

$\Rightarrow V = p U \Lambda U^T + (1-p)I$

$$= p U \Lambda U^T + (1-p) U I U^T$$

$\hookrightarrow \text{diag}(\text{eigenvalues})$ $\underbrace{\text{still}}_{=I}$

$$= U(p \Lambda + (1-p)I) U^T$$

$\Rightarrow |V| = |U(p \Lambda + (1-p)I) U^T|$

$$\Rightarrow |V| \stackrel{\text{cyclic}}{=} |p\Lambda + (1-p)\mathbb{I}|$$

$$\Rightarrow |V| = \prod_{i=1}^N \lambda_i^* \quad \lambda_i^* = \text{eigenvalues of } p\Lambda + (1-p)I$$

$$= \prod_{i=1}^N (p \lambda_i + (1-p)) \quad \lambda_i = \text{eigenvalues of } D-W$$

eigenvalues of a diagonal matrix = diagonal elements!

⊗ Deal with divergence / ϕ

$$\text{Sum}(\phi_i) \sim \mathcal{N}(0, 0.0001 \dots)$$

→ small

or

$$\begin{aligned} \text{L } \phi_{\text{raw}} &\sim \text{Car-Leroux} \\ \phi &= \phi_{\text{raw}} - \text{mean}(\phi_{\text{raw}}) \end{aligned}$$

$\Rightarrow$ Purpose: $\sum_{i=1}^N \phi_i \approx 0$ (as CARBayes)